

Finding Open Data

Finding Open Data

First, go to the ATLAS Open Data website,
<https://opendata.atlas.cern/>



Finding Open Data



The screenshot shows the ATLAS Open Data website. The navigation bar at the top includes links for 'ATLAS Open Data', 'Get Started', 'Data', 'Tutorials', 'Python API', 'Documentation', 'About us', and 'Help'. A search bar is located on the right. The main heading 'ATLAS Open Data' is prominently displayed. Below the heading is the ATLAS open data logo, which consists of a blue wireframe globe above the text 'ATLAS' and 'open data'. Two dark blue callout boxes with white text are overlaid on the image. The first callout box, pointing to the 'Data' link in the navigation bar, contains the text: 'We will click “data” to explore the data collections.' The second callout box, pointing to the 'Documentation' link, contains the text: 'You should also explore the “Documentation” tab'.

ATLAS Open Data

We will click “data” to explore the data collections.

You should also explore the “Documentation” tab

ATLAS
open data

High Energy Physics data for everyone.

Finding Open Data

The Data

The ATLAS collaboration, dedicated to advancing scientific knowledge and promoting open science, has made a significant portion of its data publicly available. This release includes two primary types of data:

- [Monte Carlo \(MC\) Simulations](#): Simulated particle collisions within the ATLAS detector.
- [Detector Data](#): Measurements recorded from particle collisions within the ATLAS detector.

These datasets provide valuable insights into high-energy particle physics and are available in two levels of complexity: for educational purposes and for research. They are good resources for researchers, educators, and students, supporting a wide range of scientific inquiries, educational projects, and exploratory analyses.

For detailed information on the data content and how to download it, explore the categories below:



Accessing the data

3 items



Data for education

3 items

Read through the information linked here.

Then click on Data for Education



Finding Open Data

Data for education

For educational data, we currently offer two different datasets: one at a center-of-mass energy of 8 TeV, corresponding to data taken in 2012, and another at 13 TeV, corresponding to data taken in 2016.

 13 TeV 2025 Data

1 item

 13 TeV 2020 Data

2 items

 8 TeV Data

1 item

We will initially be using the 2020 data.

If you want to you can explore other data sets when conducting your own research in Phase 3



Finding Open Data

🏠 > The Data > Data for education > 13 TeV 2020 Data

The 13 TeV 2020 Data

NOTE

! For detailed information about this release, you can read "[Review of the 13 TeV ATLAS Open Data release.](#)"

A set of proton-proton (pp) collision data was released by the ATLAS Collaboration to the public for educational purposes. The data has been collected by the ATLAS detector at the LHC at 13 TeV during the year 2016 and corresponds to an integrated luminosity of 10 fb^{-1} . The pp collision data is accompanied by a set of MC simulated samples describing several processes which are used to model the expected distributions of different signal and background events.

Explore the 13 TeV Data for Education

- The released samples are provided in a simplified data format, reducing the information content of the original data analysis format used within the ATLAS Collaboration.
- The resulting format is a [ROOT](#) tuple with more than 80 branches. For those not familiar with this modular scientific software toolkit, please refer to the [ROOT documentation](#), which provides a rich set of tutorials and code examples.

Read down through this page. Some parts might be confusing but you will gradually develop familiarity with the notation and the contents of the files.

You will likely revisit it and the documentation tab a few times to read about the different types of analysis you can do and to find the numbers/names of datafiles for those analyses.

When you're ready, click the blue button to navigate to the files themselves.



Finding Open Data

opendata
CERN

ATLAS 13 TeV for 2020 Open Data release

8 result(s) found

Sort by Best match

Current parameters Clear all

ATLAS Dataset

Type

- ☒ Dataset (8)
- ☐ Derived (8)
- ☐ Simulated (8)
- ☐ Documentation (1)
- ☐ About (1)
- ☐ Environment (1)
- ☐ VM (1)
- ☐ News (2)

Experiment

- ☒ ATLAS (8)

Availability

- ☐ online (8)

Year

ATLAS 13 TeV samples collection Gamma-Gamma, for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection exactly two leptons (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection exactly three leptons (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection exactly one lepton (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

ATLAS 13 TeV samples collection at least four leptons (electron or muon), for 2020 Open Data release

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. ...

Dataset Derived Simulated ATLAS

You can use filters to narrow your search. E.g. if you want Monte Carlo simulated data you can select "Derived"

Then you can click on a row to look at it in detail.

Finding Open Data

ATLAS 13 TeV samples collection exactly two leptons (electron or muon)

release

ATLAS Collaboration

Cite as: ATLAS Collaboration (2020). ATLAS 13 TeV samples collection exactly two leptons (electron or muon)
DOI:10.7483/OPENDATA.ATLAS.OHZ7.RFNH

Dataset Derived Simulated ATLAS 13TeV pp CERN-LHC

Description

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. This set of real data is accompanied by matching simulated samples of Standard Model processes and a selection of Beyond the Standard Model signals.

Dataset characteristics

210 files. 47.7 GiB in total.

How were these data generated?

Events selected with exactly two leptons (electron or muon). Both real data and simulated samples are subjected to a loose event preselection to reduce the processing time by reducing the overall number of events that have to be analysed. The preselection consists of a set of object selection criteria and a subsequent event selection acting on these preselected objects.

These DOI links are unique identifiers that can be used with the opendata client and the FileRetriever.py “print_files” method.



Finding Open Data

This dataset is provided by the ATLAS Collaboration **only for educational purposes and is not suited for scientific publications**. You can access these data through the ATLAS Virtual Machine and Software. More documentation is available at:

[The ATLAS Open Data documentation for 13 TeV release](#)

[The ATLAS Open Data site](#)

There are links to the documentation

Files and indexes

Filename	Size	
data_A.exactly2lep.root	122.1 MiB	Download
data_B.exactly2lep.root	450.8 MiB	Download
data_C.exactly2lep.root	657.0 MiB	Download
data_D.exactly2lep.root	1009.4 MiB	Download
exactly2lep.zip	20.9 GiB	Download

« ‹ 1 2 3 4 5 ... 42 › »

Sometimes there are a lot of tiles!

It is possible to download the files and work with them locally, but as they are very large this is inefficient. It also doesn't work well with the Monty server.

Finding Open Data

ATLAS 13 TeV samples collection exactly two leptons (electron or muon) Data release

ATLAS Collaboration

Cite as: ATLAS Collaboration (2020). ATLAS 13 TeV samples collection exactly two leptons (electron or muon) Data release
DOI: [10.7483/OPENDATA.ATLAS.OHZ7.RFNH](https://doi.org/10.7483/OPENDATA.ATLAS.OHZ7.RFNH)

[Dataset](#) [Derived](#) [Simulated](#) [ATLAS](#) [13TeV](#) [pp](#) [CERN-LHC](#)

Description

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. This set of real data is accompanied by matching simulated samples of Standard Model processes and a selection of Beyond the Standard Model signals.

Dataset characteristics

210 files. **47.7 GiB** in total.

How were these data generated?

Events selected with exactly two leptons (electron or muon). Both real data and simulated samples are subjected to a loose event preselection to reduce the processing time by reducing the overall number of events that have to be analysed. The preselection consists of a set of object selection criteria and a subsequent event selection acting on these preselected objects.

So instead, copy and paste this DOI link, and bring it back to your notebook.



Finding Open Data

```
#Import the uproot library
import uproot

#Here you could also import any other python libraries you would like to use
import matplotlib.pyplot as plt
import numpy as np

#Function to list datafiles from an Open Data collection
from FileRetriever import print_files as print_files
```

✓ 0.0s

In Notebook 3 we show you how to import the `print_files` method from the `FileRetriever`

```
#An example of how to print a list of the urls for e.g. https://opendata.cern/record/15005
print_files("10.7483/OPENDATA.ATLAS.2Y1T.TLGL") #Parameter = DOI from the top of the page, as a string
```

[4] ✓ 0.7s

This is how to use the DOI from the top of the page to retrieve the list of urls for the data files.

```
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep.zip
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data A.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data B.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data C.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/Data/data D.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 301215.ZPrime2000 ee.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 301216.ZPrime3000 ee.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 301220.ZPrime2000 mumu.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 301221.ZPrime3000 mumu.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 301322.ZPrime400 tt.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 301323.ZPrime500 tt.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 303329.RS G ZZ 1111 c10 m1000.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 303334.RS G ZZ 1111 c10 m2000.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 305553.Gee 01 1000.4lep.root
http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/4lep/MC/mc 305556.Gee 01 2000.4lep.root
```

Copy the link of the file that you want.

Finding Open Data

```
#f = uproot.open("")  
f = uproot.open("http://opendata.cern.ch/eos/opendata/atlas/OutreachDatasets/2020-08-19/11largeRjet1lep/Data/data\_A.11largeRjet1lep.root")
```

Then paste it into the uproot.open method as a string
(i.e. between two double quotes " ")

